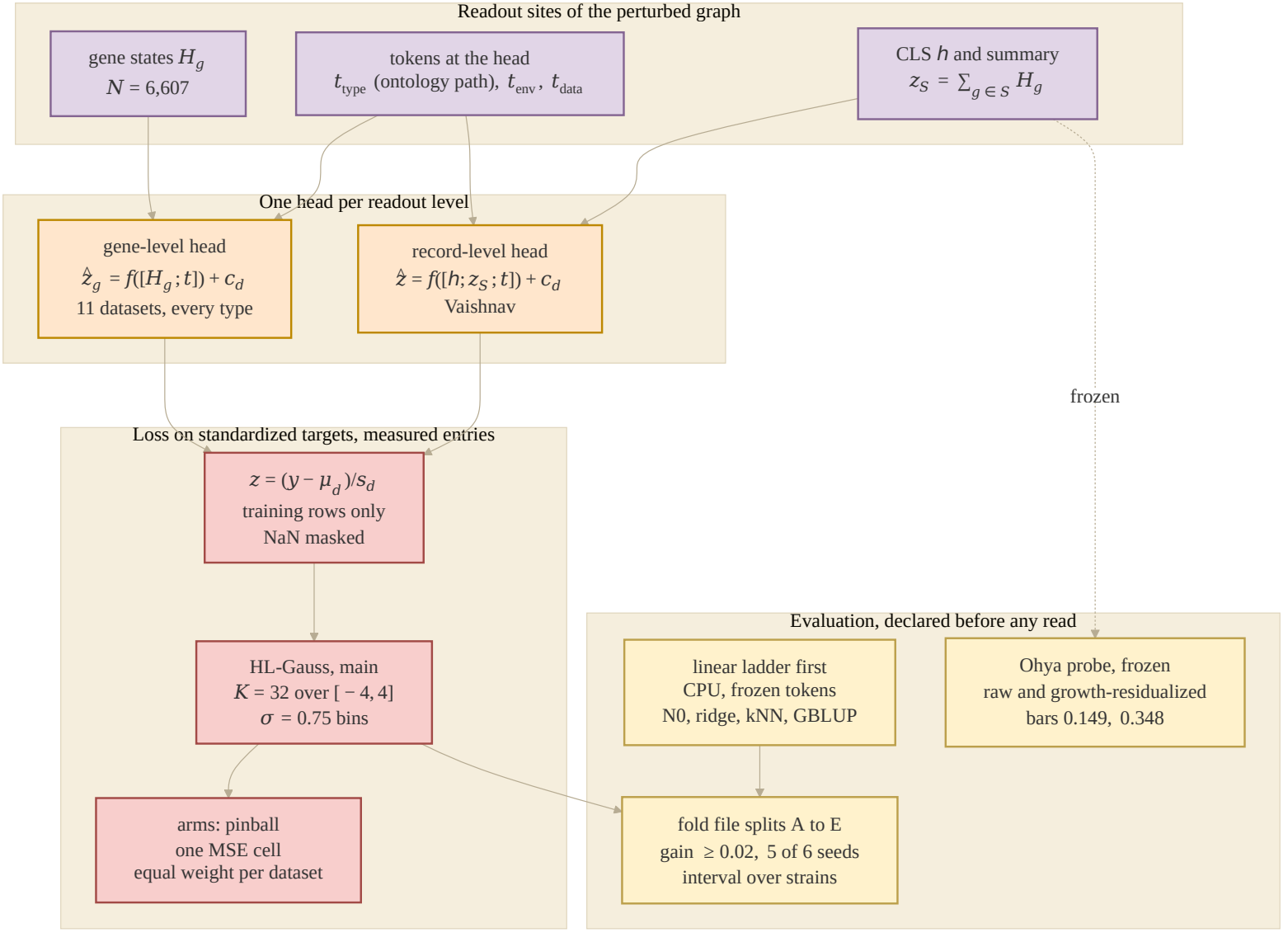

Readout sites of the perturbed graph
gene states H_g
N = 6,607
tokens at the head
t_{type} (ontology path), t_{env}, t_{data}
CLS h and summary
$z_S = \sum_{g \in S} H_g$
One head per readout level
gene-level head
$\hat{z}_g = f([H_g; t]) + c_d$
11 datasets, every type
record-level head
$\hat{z} = f([h; z_S; t]) + c_d$
Vaishnav
frozen
Loss on standardized targets, measured entries
$z = (y - \mu_d)/s_d$
training rows only
NaN masked
HL-Gauss, main
K = 32 over $[-4, 4]$
σ = 0.75 bins
arms: pinball
one MSE cell
equal weight per dataset
Evaluation, declared before any read
linear ladder first
CPU, frozen tokens
N0, ridge, kNN, GBLUP
Ohya probe, frozen
raw and growth-residualized
bars 0.149, 0.348
fold file splits A to E
gain ≥ 0.02, 5 of 6 seeds
interval over strains